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Voltage-induction type electrostatic film motor driven by two- to four-phase ac voltage and electrostatic induction

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Abstract

This paper describes a voltage-induction type electrostatic film motor that operates by feeding electric power to the slider by electrostatic induction. In the electrostatic motor, applying voltages to both of its stator and slider contributes to its fine positioning resolution and better controllability. However, in some cases such as relatively small motors, the electric wires connected to the slider could cause mechanical disturbances to the motor motion. In the new electrostatic motor, electric wires are connected only to the stator, thus the slider motion is not disturbed. The power to the slider is transferred by electrostatic induction between induction electrodes fabricated in both of the stator and the slider. The proposed motor has a two- to four-phase driving-electrode configuration, which consists of a two-phase driving electrode in the slider and a four-phase driving electrode in the stator. The paper first analyzes the thrust force characteristics of the driving-electrode configuration, and then analyzes the characteristics of voltage-induction configuration, both based on capacitance-network analysis. Experimental results showed good agreements with the analyses, confirming the feasibility of the voltage-induction type electrostatic motor.
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Keywords: Electrostatic actuator; Film actuator; Synchronous motor; Electrostatic induction; Flexible printed circuit

1. Introduction

In advanced mechatronic systems, such as robots and advanced industrial equipments, new actuators with higher performance or with special features have been highly expected. For example, in the robotic field, lightweight and high-power actuators, which are utilized for artificial muscle, have been demanded to enhance dynamic performance of robots. On the other hand, in some industrial or scientific equipments, such as electron beam (EB) lithography system and magnetic resonance imaging (MRI) scanners, nonmagnetic actuators are required that do not interfere with their functions [1,2]. To fulfill those requirements, various new actuators such as pneumatic actuators, hydraulic actuators, ultrasonic actuators, shape-memory-alloy actuators (SMA), ionic conducting polymer gel film actuators (ICPF), and electrostatic actuators have been studied [3–9].

A high-power electrostatic motor is one of promising actuators for above-mentioned fields. Electrostatic motors were originally well-known in the field of micro-machines because of its high-power density in microregion. Various types of electrostatic motors such as stepping motor, synchronous motor, comb drive, wobble motor, scratch drive, inch worm motor, have been studied using MEMS technology [10–21]. On the other hand, recent studies have revealed that even centimeter-sized electrostatic actuators can possess high-power density by patterning fine electrodes on a large plastic film using the flexible printed circuit (FPC) technology [8,9,22], which are called electrostatic film motors. For example, the film motor called *dual excitation multiphase electrostatic drive* (DEMED) showed superior output performance (e.g. power density of 230 W/kg, maximum speed of over 1 m/s) and some unique characteristics, such as mechanical flexibility, and thus are promising for various mechatronic applications [23,24].

From the viewpoint of power feeding, electrostatic film motors studied before can be roughly classified into two types. In one type, electric power is supplied to both stator and slider [9,22]. In the other type, electric power is supplied only to stator [8]. It can be speculated from the comparison of those two types

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that supplying power to both the stator and the slider results in better controllability in terms of positioning resolution. For example, the DEMED achieved fine positioning resolution [25] by applying ac high voltages on both stator and slider. However, electric wires used for power feeding to the slider could sometimes cause mechanical disturbances to the slider motion (especially in the case of relatively small motors). In the other type that has no wiring to slider, the slider motion is free from the mechanical disturbances and therefore some unique applications have been proposed [8,26]. However, its positioning resolution is limited by the electrode pitch.

In this research, we propose and evaluate a new type of high-power electrostatic film motor, in which electric power is fed to both films but the power to the slider is fed by electrostatic induction. The electrostatic induction frees the slider from cables yet realizing better controllability and finer positioning resolution than the previous wireless-slider type. Since the driving-electrode configuration is different from previously reported high-power motors, we will first analyze thrust force characteristics of the driving-electrode configuration by ignoring the effect of electrostatic induction in Section 3. Then, we discuss the effect of the induction and reveal the total performance of the newly proposed motor in Sections 5 through 7.

2. Voltage-induction type electrostatic motor

Fig. 1 shows a schematic diagram of the voltage-induction type electrostatic motor proposed in this paper. The motor consists of a pair of thin plastic films, a slider and a stator, fabricated by flexible print circuit (FPC) board technology. Each of the films involves driving electrodes and induction electrodes. A four-phase parallel driving electrode is located at the center of the stator, whereas a two-phase parallel driving electrode is located at the center of the slider. Both the stator and the slider have two equally sized induction electrodes at both sides of the driving electrodes. In the slider, driving electrodes and induction electrodes are integrated, respectively, to conform an inter-digital electrode. In the stator, driving electrode and induction electrode are separated. Electric power to the stator is directly supplied to the stator's driving electrode from four-phase ac power supply through electric cables. On the other hand, the power to the slider is first supplied to the stator's induction electrode from two-phase ac power supply, then it is transferred to the slider's induction electrodes by electrostatic induction, simultaneously to the integrated driving electrodes. Thus, no direct cabling to the slider is required. Stacking a slider on a stator and applying voltages only on the stator film, we can obtain a linear motion perpendicular to the parallel electrode.

In this motor, the number of induction electrodes must be the same number of phases in slider's driving electrode. Since lower number of induction electrodes decreases total induction-electrode area, the phases of slider's driving electrode should preferably be minimized. Therefore, we employed two-phase driving electrode for the slider. Two-phase driving electrodes have another advantage, which is low manufacturing cost. When manufacturing FPC films, three-phase or four-phase electrode requires double-sided print board, whereas two-phase electrode can be realized by single-sided print board that is cheaper than double-sided.

The number of phases in stator's driving electrodes must be more than three when we choose two-phase electrode for the slider. In this work, we choose four-phase electrode for the stator to simplify the driving-voltage generation, that is, we can easily generate two voltages by reversing the other two voltages using transformers. In the resultant two- to four-phase configuration, the stator electrodes have the regular electrode pitch p, whereas the slider electrodes have pitch twice as large as that of the stator.

3. Force and speed analysis of two- to four-phase electrostatic motor

The purpose of this paper is to provide analytical and experimental evaluations about voltage-induction type electrostatic motor with two- to four-phase driving electrode. Since the two- to four-phase driving-electrode configuration is different from the previously studied three- to three-phase one, first, we evaluate thrust force characteristics of its new driving-electrode configuration. In the analysis, we ignore the electrostatic induction, that is, we assume the condition that power supply to the two- to four-phase electrode is directly fed by electric cables. The total characteristics including the effect of electrostatic induction is discussed in the latter part of the paper. The motor is modeled as a capacitance network, with which thrust force is formulated. Then, we discuss the speed of the motor and propose suitable driving method of the motor.

Fig. 2 shows the schematic diagram of the two- to four-phase electrostatic motor (without induction). This section analyzes the force characteristics and speed of the motor in Fig. 2 based on

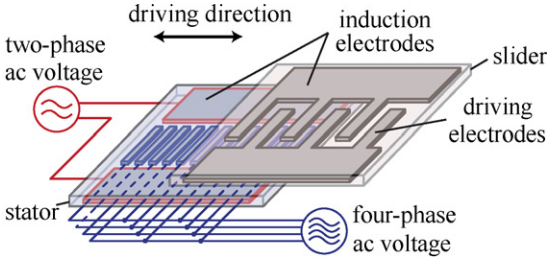


Fig. 1. Voltage-induction type electrostatic motor.

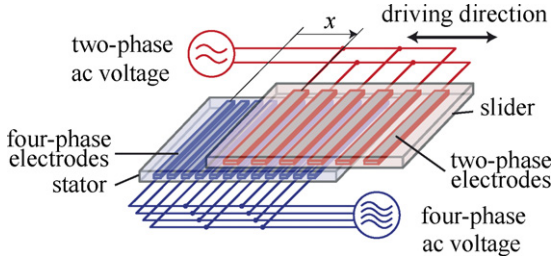


Fig. 2. Two- to four-phase electrostatic motor.

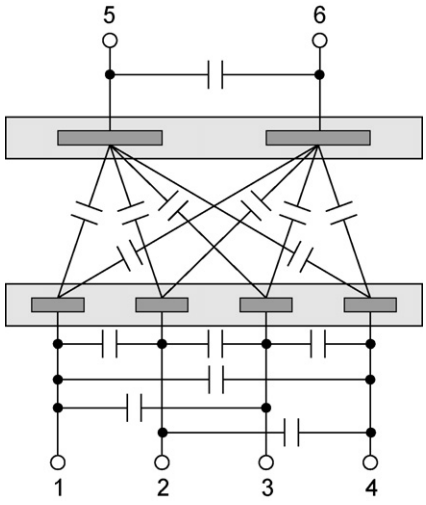


Fig. 3. Six-terminal model of two- to four-phase motor.

the method described in [25]. In an electrostatic motor, the relation among electrodes can be represented as capacitors. Since the two- to four-phase electrostatic motor has six phases in total, it can be modeled as a six-terminal capacitance network. The capacitance network model of the motor is illustrated in Fig. 3. These capacitances can be mathematically represented using a 6×6 capacitance-coefficient matrix. In the matrix, element in ith row and jth column expresses the relationship between ith terminal and jth terminal. In the two- to four-phase motor, considering the geometrical symmetry among electrodes, the capacitance matrix can be written as

$$\mathbf{C} = \begin{bmatrix} C_{\mathrm{st}} & -C_{\mathrm{ta}} & -C_{\mathrm{tb}} & -C_{\mathrm{ta}} & C_{\mathrm{m}}(\theta_x) & C_{\mathrm{m}}(\theta_x - \pi) \\ -C_{\mathrm{ta}} & C_{\mathrm{st}} & -C_{\mathrm{ta}} & -C_{\mathrm{tb}} & C_{\mathrm{m}}\left(\theta_x - \frac{\pi}{2}\right) & C_{\mathrm{m}}\left(\theta_x + \frac{\pi}{2}\right) \\ -C_{\mathrm{tb}} & -C_{\mathrm{ta}} & C_{\mathrm{st}} & -C_{\mathrm{ta}} & C_{\mathrm{m}}(\theta_x - \pi) & C_{\mathrm{m}}(\theta_x) \\ -C_{\mathrm{ta}} & -C_{\mathrm{tb}} & -C_{\mathrm{ta}} & C_{\mathrm{st}} & C_{\mathrm{m}}\left(\theta_x + \frac{\pi}{2}\right) & C_{\mathrm{m}}\left(\theta_x - \frac{\pi}{2}\right) \\ C_{\mathrm{ma}}(\theta_x) & C_{\mathrm{m}}\left(\theta_x - \frac{\pi}{2}\right) & C_{\mathrm{m}}(\theta_x - \pi) & C_{\mathrm{m}}\left(\theta_x + \frac{\pi}{2}\right) & C_{\mathrm{sl}} & -C_{\mathrm{l}} \\ C_{\mathrm{m}}(\theta_x - \pi) & C_{\mathrm{m}}\left(\theta_x + \frac{\pi}{2}\right) & C_{\mathrm{m}}(\theta_x) & C_{\mathrm{m}}\left(\theta_x - \frac{\pi}{2}\right) & -C_{\mathrm{l}} & C_{\mathrm{sl}} \end{bmatrix} \tag{1}$$

$$C_{\mathrm{m}}(\theta_x) = -C_{\mathrm{m0}} - C_{\mathrm{m1}} \cos\theta_x \tag{2}$$

$$\theta_x = \frac{\pi x}{2p} \tag{3}$$

where C_{st}, C_{sl}, C_{ta}, C_{tb}, C_l, C_{m0} and C_{m1} are positive coefficients, x is the relative position of the slider electrodes and θ_x is the another representation of the position x in electric angle, which one cycle (2π) is equal to $4p$. The coefficients of the matrix are determined by geometric relations among electrodes, and in practice, they can be measured using an LCR meter [25]. Assuming charge and voltage vectors, the capacitance-coefficient matrix satisfies the following condition:

$$\mathbf{q} = \mathbf{CV} \tag{4}$$

where $\mathbf{q}$ and $\mathbf{V}$ are 6×1 vectors that represent the charges and voltages on the six terminals, respectively. In the case of Fig. 2, $\mathbf{V}$ is expressed as follows:

$$\mathbf{V} = \{V_t \sin\omega_t t, -V_t \cos\omega_t t, -V_t \sin\omega_t t, V_t \cos\omega_t t, V_l \sin\omega_l t, -V_l \sin\omega_l t\}^{\mathrm{T}} \tag{5}$$

where V_t, V_l, ω_t, ω_l are voltage amplitudes and angular frequencies of stator and slider voltages.

Based on the principle of virtual work, thrust force of the motor can be derived from a derivative by x of electrostatic energy conserved in the capacitances. Using the coefficient matrix and the voltage vector, thrust force can be obtained as

$$f_x = \frac{1}{2}\mathbf{V}^{\mathrm{T}}\frac{\partial \mathbf{C}}{\partial x}\mathbf{V} \tag{6}$$

Substituting for (1) and (5), we can rewrite (6) as

$$f_x = \frac{\pi C_{m1}}{p} V_l V_t \{\sin(\omega_l t + \omega_t t - \theta_x) + \sin(\omega_l t - \omega_t t + \theta_x)\} \tag{7}$$

This formula shows that the thrust force contains two sinusoidal components, whose frequencies are the sum of frequencies and the difference of frequencies, while the previous electrostatic motor, which has three-phase electrode in both films, has only one sinusoidal component [25].

We attribute these components to the slider's standing voltage distributions (waves) that is derived from the two-phase electrode configuration. If, for example, we have three-phase electrodes both on slider and stator to which three-phase ac voltages are applied, then we have two traveling voltage distributions (waves), respectively, on statorand on slider. In this case, thrust force is represented using thrust force coefficient κ as follows:

$$f_l = \kappa V_l V_t \sin\left(\omega_l t - \omega_t t + \frac{2\pi}{3p}x\right) \tag{8}$$

When there is a constant load f_l, then the following condition should satisfy:

$$\sin\left(\omega_l t - \omega_t t + \frac{2\pi}{3p}x\right) = \frac{f_l}{\kappa V_l V_t} = \text{const.} \tag{9}$$

$$x_{\mathrm{three}} = \frac{3p}{2\pi}(\omega_t t - \omega_l t + \alpha) \tag{10}$$

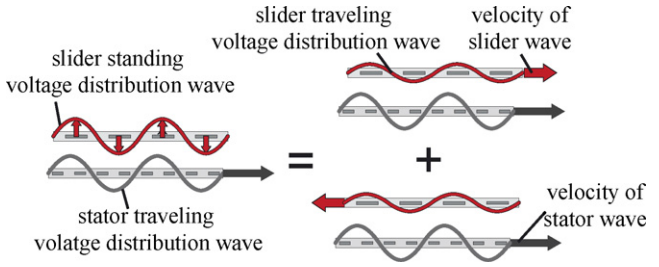


Fig. 4. Slider standing voltage distribution wave is regarded as superposition of two traveling electric field wave.

where α represents a constant value. Then, the speed of the motor becomes as follows:

$$u_{\text{three}} = \frac{\partial x_{\text{three}}}{\partial t} = \frac{3p}{2\pi}(\omega_t - \omega_l) \tag{11}$$

This means that the three- to three-phase motor runs at the speed difference between the two traveling waves.

On the contrary, in the case of two-phase electrode, we cannot excite traveling voltage wave on the slider. Applying two-phase voltage generates standing voltage wave as in Fig. 4. Mathematically, a standing wave can be regarded as superposition of two traveling waves, which run toward opposite direction to each other with half amplitude of the original standing wave. Therefore, in the case of two- to four-phase electrode, interactions between the traveling wave of the stator and the each of two traveling waves of the slider generate two different thrust force components; each component tries to drive the slider with the speed difference between the corresponding slider wave and the stator wave. The two components in (7) are representing those two forces. There is no simple solution for the speed as in (11).

Here, we propose to virtually eliminate one component by using high frequency ac voltages. That is, using high and close frequencies for ω_t and ω_l, the first term in (7), which is $\sin(\omega_l t + \omega_t t - \theta_x)$ becomes too high frequency that the motor cannot mechanically respond.

Therefore, it can be neglected and the thrust force in (7) can be simplified to

$$f_x = \frac{\pi C_{m1}}{p} V_l V_t \sin(\omega_l t - \omega_t t + \theta_x) \tag{12}$$

As the same manner to derive (11), the speed of the motor becomes as follows:

$$u = \frac{2p(\omega_t - \omega_l)}{\pi} \tag{13}$$

This analysis reveals that when the motor is driven with high frequencies, such as 1 kHz, smooth motion characteristic is obtained. In addition, using high frequency has another advantage to make the voltage supply smaller because high voltage with higher frequency can be easily generated by smaller transformers.

4. Experimental verification

We measured thrust force of the two- to four-phase electrostatic motor in two condition to verify the analysis. It should be noted that the motor verified in this section is not a voltage-induction type motor but a motor shown in Fig. 2, which applies voltages to slider directly using electric wires. In the first measurement, low-frequency voltages are used for driving voltage, and in the second measurement, high-frequency voltages are used. The slider and stator films used for the measurement are shown in Fig. 5. Both the slider and the stator are fabricated by FPC technology. The effective electrode of both films have same-sized areas, which measure 30-mm wide by 46-mm long. The electrodes pitch p of the stator is 200 μm and that of the slider is 400 μm.

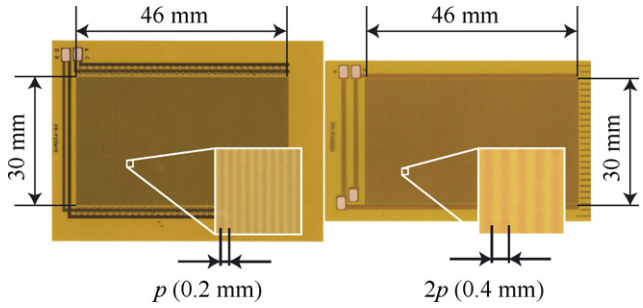


Fig. 5. Stator (left) and slider (right) films. Both films have parallel electrodes inside them.

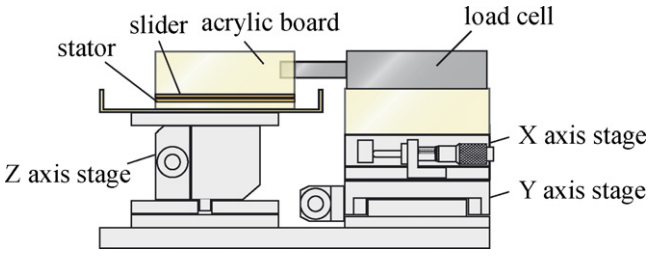


Fig. 6. Experimental setup for force measurement.

4.1. Thrust force at low-frequency voltages

Fig. 6 shows the experimental setup for force measurement. The slider is glued to an acrylic board, which is attached to a load-cell, and is placed on the stator fixed to the base. Fig. 7 shows the equipment configuration of generating driving voltages. Two-phase function generator (WF1946, NF circuit block) and four-phase function generator (AG1200, Yokogawa) generate two-

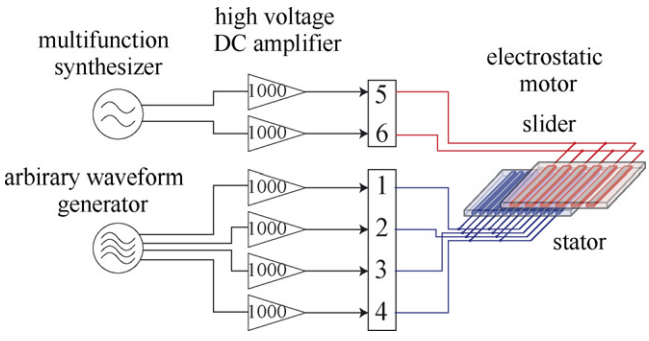


Fig. 7. Low-frequency voltage supply.

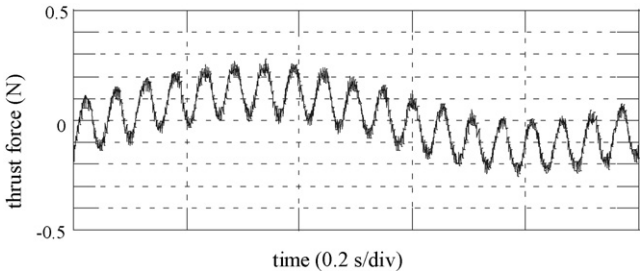


Fig. 8. Thrust force of the motor driven by low-frequency voltage (stator-voltage frequency of 10 Hz and slider-voltage frequency of 9 Hz). The force contained two-frequency component (1 Hz and 19 Hz). Thrust force was measured fixing both slider and stator to keep at the same position.

phase and four-phase sinusoidal waves, which are amplified with a gain of 1000 by high voltage amplifiers (model 609C, Trek) before being fed to the electrodes. Although the applied voltages should be higher than 500 V_{0-p} to obtain a practically large force, such high voltages can cause electric discharge between the two films. To prevent the discharge, the motor is immersed in dielectric liquid (Fluorinert FC-77, 3M). Furthermore, glass beads of 10- μm diameter are scattered between the two films to reduce the friction between the slider and the stator.

Fig. 8 shows a measured thrust force in the condition that the stator's four-phase voltage was 1 kV_{0-p} with 10 Hz and the slider's two-phase voltage was 1 kV_{0-p} with 9 Hz. It should be reminded that, in this measurement, the slider was fixed so θ_x in (7) is a certain constant value θ_0. That is, measured thrust force in Fig. 8 corresponds to the following equation:

$$f_x = \kappa_0\{\sin(\omega_l t + \omega_t t - \beta) + \sin(\omega_l t - \omega_t t + \beta)\} \quad (14)$$

where, $\kappa_0 = \pi C_{m1} V_l V_t / p$. In this measurement, since θ_0, relative position of slider/stator, only shifts the phase of the thrust force characteristic line, we can obtain almost the same results at any slider positions. The observed two frequencies in Fig. 8 correspond to the two components in (14). In the plot of Fig. 8, the components of $\omega_l - \omega_t$ (i.e., 1 Hz) and $\omega_l + \omega_t$ (i.e., 19 Hz) can be clearly observed, which verifies (7).

Fast Fourier transformation (FFT) analysis of the plot in Fig. 8 reported that the amplitude of 1 Hz component was 0.1 N and of 19 Hz was 0.085 N. Since the measured value of C_{m1} in the setup was 8.8 pF, the theoretical amplitude of both components is 0.14 N. Although the measured amplitudes, especially for 19 Hz component, are smaller than the theoretical value, it would be due to mechanical attenuation in the force measurement setup.

4.2. Experiment of measuring thrust force at high-frequency voltages

Next, to verify that the higher frequency components (sum of two frequencies) in (7) can be ignored by using high frequency signals, we measured thrust force with higher ac frequencies. To generate high frequency high-voltages, a combination of high-speed amplifiers (HSA4012, NF circuit block, voltage gain: 10) and transformers that have 1:33 turns ratio are utilized as shown in Fig. 9.

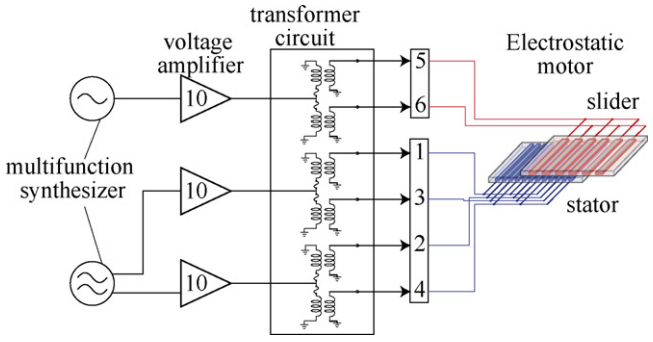


Fig. 9. High-frequency voltage supply.

Fig. 10 shows the measured thrust force under the condition that stator voltage was 1 kV_{0-p} with 1000 Hz, and slider voltage was 1 kV_{0-p} with 999 Hz. The plot clearly shows the component of $\omega_l - \omega_t$ (1 Hz). The component of $\omega_l + \omega_t$ (1999 Hz) looks completely attenuated. In this case, the high frequency component would have been attenuated by the limited frequency range of the force measurement setup. However, in normal operation, the inertia of the slider would give the same attenuation effect. Then, thrust force can be assumed as Fig. 10. This speculation is verified in the next subsection.

In this setting, the measured capacitance of C_{m1} was 19.4 pF (The value is different from the measurement in Fig. 8. This is because the fixing condition of the slider in the experimental setup can be different at each time). Theoretical thrust force amplitude of the frequency-difference component is 0.305 N. This value agrees quite well with the experimental result.

4.3. Motor speed verification

Next, we operated the motor by the high-frequency voltage setting. The slider with electric wire was simply stacked on the stator as shown in Fig. 11, and then driven by voltage excitation. In the operation, stator frequency was fixed at 1000 Hz, and slider frequency was switched between two value, 999 and 1001 Hz, to obtain reciprocating motion. In addition, we selected another slider frequency pair, 998 and 1002 Hz, to confirm that the slider speed changes in proportion to the difference of frequencies. The motion of the slider was measured by an optical displacement sensor (Zimmer, 200) as shown in Fig. 12. The measured motion is plotted in Fig. 13. The slider runs with

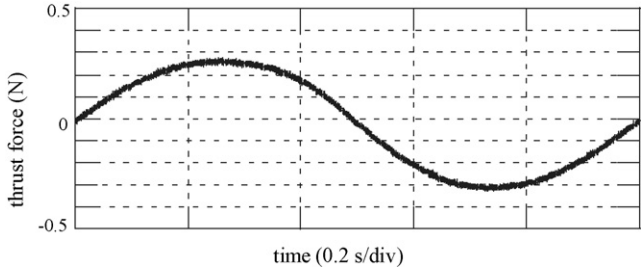


Fig. 10. Thrust force of the motor driven by high-frequency voltage (stator-voltage frequency of 1000 Hz and slider-voltage frequency of 999 Hz). The force contained only one frequency component (1 Hz). Thrust force was measured fixing both slider and stator to keep at the same position.

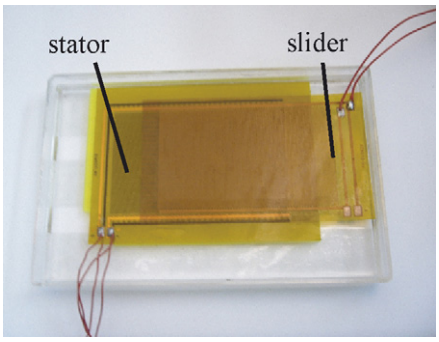


Fig. 11. Configuration of slider and stator in driving. The slider is simply stacked on the stator.

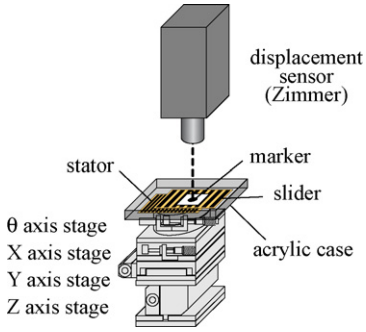


Fig. 12. Equipment for measuring displacement of the motor by optical sensor.

a speed of 0.80 mm/s in the condition that slider frequencies are 999/1001 Hz, and 1.60 mm/s in the condition that slider frequencies are 998/1002 Hz. These speeds agree well with the values that theoretical analysis predicted, and apparently they were proportional to the frequency difference of slider and stator.

4.4. Component of sum of frequencies in motor motion

Finally, we evaluated the vibration that is caused by the thrust force with sum of frequencies in (7). Although we achieved smooth motion of the motor in the previous subsection by virtually eliminating the thrust force component with sum of frequencies, in fact, the motor was slightly vibrating when it

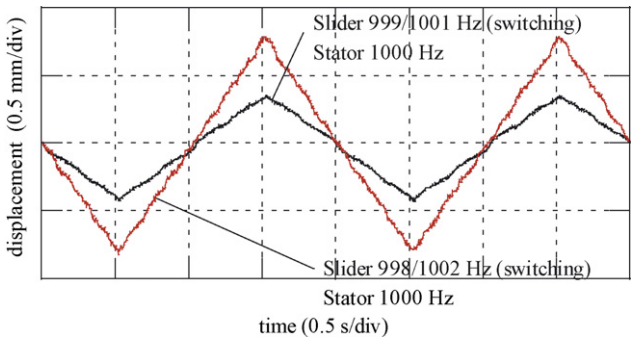


Fig. 13. Slider motion at the condition of high-frequency voltage (stator-voltage frequency of 1000 Hz and slider-voltage frequency of 999 Hz).

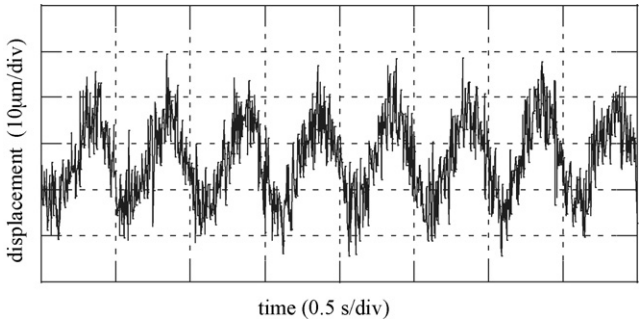


Fig. 14. Vibration caused by the thrust force component of sum of frequencies.

was moving because the component was not completely attenuated. For example, when we set the voltage frequencies for slider and stator at the same value of 1000 Hz, the motor motion should stop under the assumption, but the slider is still vibrating at the sum of frequencies (i.e 2000 Hz) as shown in Fig. 14. We can diminish the amplitude of the vibration by raising the frequencies of driving voltages, which results in high-frequency force. The frequency characteristic of the vibrating components is plotted in Fig. 15. It is observed that the amplitude simply decreases and no resonant frequency exists in the result. The vibration was suppressed lower than 10 μm in higher frequency condition. It can be lowered also by attaching a load to the motor. In the practical application, the driving frequency should be decided by the acceptable amplitude of the vibration on each application.

5. Analysis of voltage-induction type two- to four-phase electrostatic motor

In this section, we discuss the effect of electrostatic induction. Voltage-induction type two- to four-phase electrostatic motor is also modeled by a capacitance network model. Using the model, we formulate induction voltage and thrust force of the motor. Furthermore, the optimized ratio of driving-electrode area against total area is calculated.

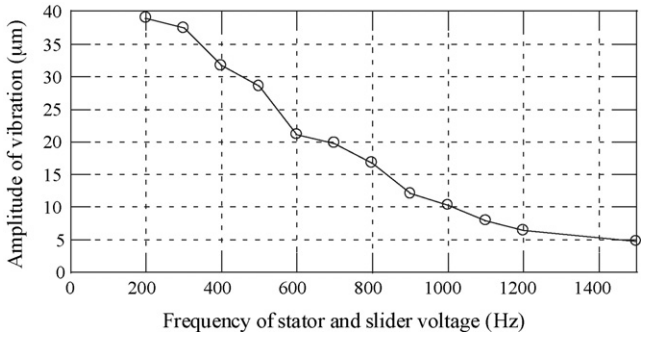


Fig. 15. Frequency characteristic of the amplitude of vibration. In a simple rigid body model, the vibration amplitude is inversely proportional to the to square of thrust force frequency according to the Laplace transform of dynamic equation. Although actual motor model would be more complicated, the measured result shows that the vibration amplitude simply decreases with respect to the voltage-supply frequency, which links with force frequency.

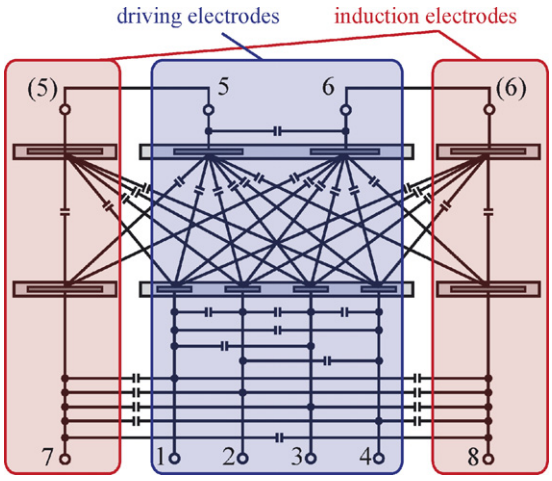


Fig. 16. Six-terminal model of voltage-induction type two- to four-phase electrostatic motor.

5.1. Motor model with induction

Fig. 16 shows a capacitance network model of the voltage-induction type motor. In this model, the relations of induction electrodes as well as that of driving electrodes are modeled by capacitance simultaneously. Although capacitances between induction electrodes and driving electrodes exists as shown in Fig. 16, we ignore them for simplicity and assume the model as shown in Fig. 17. This model is valid if the electric field is enclosed in each electrode part and does not interact between the electrode parts. The analysis based on the model in Fig. 16 is also conducted in "Appendix A".

In the model of Fig. 17, we can assume the induction electrodes as simple capacitors inserted between a voltage supply and the driving electrodes of the slider. Driving electrode part is the same model as Fig. 3. For ease of calculation, intermediate terminals No. 5 and No. 6 are defined as in Fig. 17.

With the intermediate terminals, the relationship of the slider and the stator is represented as an 8×8 capacitance-coefficient matrix. The total capacitance matrix $\mathbf{C}_{\text{total}}$ can be decomposed into two different parts, $\mathbf{C}_{\text{ind}}$ and $\mathbf{C}_{\text{drv}}$, which represent the relationship of the induction electrodes and that of the driving electrodes, respectively. These two capacitance matrices are represented asfollows:

$$\mathbf{C}_{\text{ind}} = \left[\begin{array}{c|cccc} \mathbf{O}_4 & \multicolumn{4}{c}{\mathbf{O}_4} \\ \hline & C_{\text{i}} & 0 & -C_{\text{i}} & 0 \\ \mathbf{O}_4 & 0 & C_{\text{i}} & 0 & -C_{\text{i}} \\ & -C_{\text{i}} & 0 & C_{\text{i}} & 0 \\ & 0 & -C_{\text{i}} & 0 & C_{\text{i}} \end{array}\right] \tag{15}$$

$$\mathbf{C}_{\text{drv}} = \left[\begin{array}{c|c} \mathbf{C} & \mathbf{O}_{62} \\ \hline \mathbf{O}_{26} & \mathbf{O}_2 \end{array}\right] \tag{16}$$

C_i in (15) is the capacitances between a slider induction electrode and a corresponding stator induction electrode. $\mathbf{C}$ in (16) is same as $\mathbf{C}(\theta_x)$ in (1).

5.2. Induced voltage analysis

For the analysis, we represent the unknown voltage of terminals five and six as V_5 and V_6, respectively:

$$\mathbf{V}_I = \{V_t \sin\omega_t t, -V_t \cos\omega_t t, -V_t \sin\omega_t t, V_t \cos\omega_t t, V_5, V_6, V_l \sin\omega_l t, -V_l \sin\omega_l t\}^{\text{T}} \tag{17}$$

Then, $\mathbf{C}_{\text{ind}}$, $\mathbf{C}_{\text{drv}}$ and $\mathbf{V}_I$ satisfies the following:

$$\mathbf{q}_I = (\mathbf{C}_{\text{ind}} + \mathbf{C}_{\text{drv}})\mathbf{V}_I \tag{18}$$

where charge vector is represented as follows:

$$\mathbf{q}_I = \{q_1, q_2, q_3, q_4, 0, 0, q_7, q_8\}^{\text{T}} \tag{19}$$

Since fifth and sixth terminals do not have connections to any power source, the total charges of those terminals remain zero. Substituting (15)–(17), and (19) for (18) derives the induced voltages on fifth and sixth terminals as follows:

$$V_5 = \frac{C_i V_l \sin\omega_l t + 2C_{m1} V_t \sin(\omega_t t - \theta_x)}{C_i + C_l + C_{\text{sl}}}, \qquad V_6 = -V_5 \tag{20}$$

5.3. Thrust force analysis

With the induced voltages, thrust force f_{ind} is derived using principle of virtual work as

$$f_{\text{ind}} = \frac{1}{2}\mathbf{V}_I^{\text{T}} \frac{\partial(\mathbf{C}_{\text{ind}} + \mathbf{C}_{\text{drv}})}{\partial x}\mathbf{V}_I + \mathbf{V}_I^{\text{T}}(\mathbf{C}_{\text{ind}} + \mathbf{C}_{\text{drv}})\frac{\partial \mathbf{V}_I}{\partial x} \tag{21}$$

Compared with (6), (21) has the additional second term, which arose because the induced voltages are functions of x. From (15)–(17), (20), and (21), the following thrust force formula is derived:

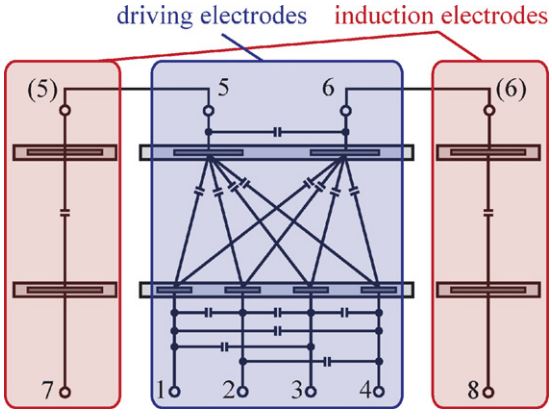


Fig. 17. Six-terminal simplified model of voltage-induction type two- to four-phase electrostatic motor.

$$f_{\text{ind}} = \frac{\pi C_{m1} V_t}{(C_i + C_l + C_{\text{sl}})p}\{2C_{m1}V_t \sin(2\omega_t t - 2\theta_x) + C_i V_l \sin(\omega_l t + \omega_t t - \theta_x) + C_i V_l \sin(\omega_l t - \omega_t t + \theta_x)\} \quad (22)$$

This formula suggests that the thrust force contains three frequency components: the double frequency of stator driving voltage, the sum of frequencies, and the difference of frequencies. Now the same discussion as the Section 3 can be applied to this thrust force equation. If ω_t and ω_l are high enough, the frequencies of the first and the second terms in (22) becomes so high that motor cannot respond. Then, the thrust force equation can be simplified to

$$f_{\text{ind}} = \frac{\pi C_{m1} C_i V_l V_t}{(C_i + C_l + C_{\text{sl}})p} \sin(\omega_l t - \omega_t t + \theta_x) \quad (23)$$

This equation is quite similar to (12), although the force magnitude is lowered.

5.4. Optimization of driving-electrode ratio

In this subsection, we discuss the optimized driving–area ratio, which maximize the thrust force. We define S_{ratio}, which is the ratio of the driving electrode in whole electrodes area. Then, following relations are obtained:

$$S_{\text{drv}} = S_{\text{ratio}} \cdot S \quad (24)$$

$$S_{\text{ind}} = (1 - S_{\text{ratio}})S \quad (25)$$

where S_{drv}, S_{ind}, and S are driving-electrode area, induction-electrode area, and whole electrodes area of the motor, respectively. In addition, we assume that the capacitance fraction of C_{sl} and C_i per unit area is k: 1. That is,

$$\frac{C_{\text{sl}}}{S_{\text{drv}}} : \frac{C_i}{S_{\text{ind}}/2} = k : 1 \Leftrightarrow S_{\text{ind}} = S_{\text{drv}} \cdot \frac{2kC_i}{C_{\text{sl}}} \quad (26)$$

From (24)–(26), following relationship is obtained:

$$C_i = \frac{C_{\text{sl}}}{2k} \cdot \frac{S_{\text{ind}}}{S_{\text{drv}}} = \frac{C_{\text{sl}}(1 - S_{\text{ratio}})}{2k \cdot S_{\text{ratio}}} \quad (27)$$

Thrust force per unit area is derived from (23), (24), and (27) by removing S and C_i as follows:

$$f_{\text{unit}} = \frac{f_{\text{ind}}}{S} = \frac{\pi C_{m1} C_{\text{sl}} V_l V_t S_{\text{ratio}}(1 - S_{\text{ratio}}) \sin(\omega_l t - \omega_t t + \theta_x)}{pS_{\text{drv}}\{C_{\text{sl}} + (2kC_l - C_{\text{sl}} + 2kC_{\text{sl}})S_{\text{ratio}}\}} \quad (28)$$

From the differentiation of (28) by S_{ratio}, optimized S_{ratio}, which maximize the f_{unit}, is represented as follows:

$$S_{\text{opt_ratio}} = \frac{-C_{\text{sl}} + \sqrt{2kC_{\text{sl}}(C_l + C_{\text{sl}})}}{2kC_l + (2k - 1)C_{\text{sl}}} \quad (29)$$

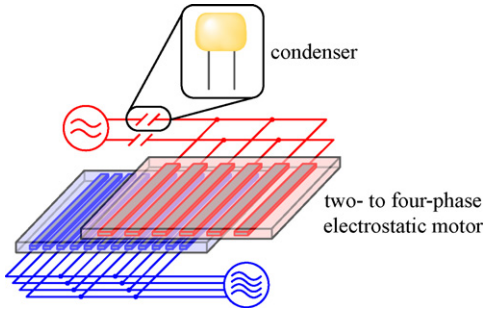


Fig. 18. Experimental model using two- to four-phase electrostatic motor and inserted condensers.

6. Experimental verification of induction based on the motor model

In this section, we conduct experiments with a motor model, which consists of previous-mentioned two- to four-phase motor and condensers, to investigate induced voltage, thrust force, and optimized driving-electrode–area ratio.

6.1. Experimental setup

We modeled a voltage-induction type motor as shown in Fig. 18.

Two- to four-phase electrostatic film motor shown in Fig. 5 is assumed as a driving-electrode and normal condensers are substituted for induction electrodes. A condenser is inserted between a two-phase power supply and a driving electrode of a slider. Condensers with different capacitances are utilized to model different sizes of induction electrodes. We utilized the same signal generators, amplifiers, and transformers of Section 4.2.

6.2. Induced voltage

The measurements of induced voltages were conducted under several conditions of S_{ratio}. These several conditions of S_{ratio} are realized by changing C_i, which is the capacitance of the connected condenser and is reduced to induction electrode area. In the utilized motor, measured capacitances are as follows:

$$C_{m1} = 19.4\,\text{pF}, \quad C_l = 123.2\,\text{pF}, \quad C_{\text{sl}} = 249.1\,\text{pF} \quad (30)$$

Using (26), (30), $S_{\text{drv}} = 13.8\,\text{cm}^2$, and assumption of $k = 1$, we could regard 1 pF of connected capacitor as $0.111\,\text{cm}^2$ of induction electrodes.

The voltage conditions are as follows: the four-phase voltage to the stator driving electrodes is $1\,\text{kV}_{0-p}$ and 1000 Hz, and the two-phase voltage to the assumed induction electrodes is $1\,\text{kV}_{0-p}$ and 999 Hz.

The measured induced voltages are plotted in Fig. 19 against S_{ratio} with theoretically calculated values using (20). Since the induced voltage has two components with frequencies of ω_t and ω_l as shown in (20), we plotted the two decomposed voltages (999 and 1000 Hz components) as well as total value. These experimental results are considerably in good agreement with the theoretical line.

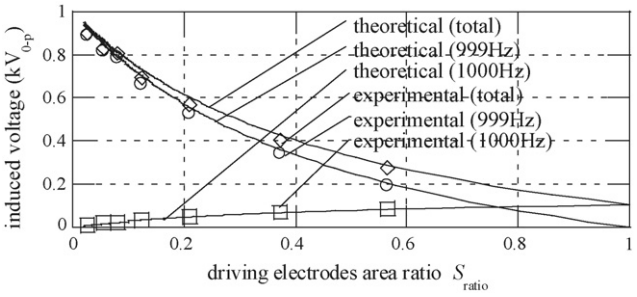


Fig. 19. Induced voltages are show in this graph. The frequency condition of applied voltages to the driving electrodes of the stator and those to the induction electrodes of the stator are 1000 and 999 Hz, respectively. The total voltage and its each element, which has the frequency of 999 and 1000 Hz are shown, respectively.

6.3. Thrust force

Fig. 20 shows the thrust forces in the voltage conditions mentioned above. For this measurement, the slider was fixed to a load-cell as same as in the measurement in Section 4. Thrust force was measured in several conditions of C_i. It was found that the amplitude of the thrust force increases according to the increase of C_i. Besides, in each thrust force, we can observe that the first and the second terms (high-speed terms) in (22) were completely attenuated. The result matches the simple thrust force characteristics obtained as (23).

6.4. Optimized driving-electrode–area ratio

Next, thrust forces of the motor against the S_{ratio} are also plotted in Fig. 21. These S_{ratio} are calculated by (27) using C_i, C_{sl} in (30), and $k = 1$. Theoretical thrust line is obtained from (28) by substituting $V_l = V_t = 1000$ V, $p = 200 \times 10^{-6}$ m, $S_{drv} = 13.8\,cm^2$, and $k = 1$. We can observe that the optimum S_{ratio}, at which the thrust force per unit area becomes the largest, is in the range of $S_{ratio} = 0.3$–0.4 in both theoretical and experimental results.

The experimental amplitude of the thrust force was smaller than that of the theoretical result. The possible reasons of the disagreement are friction, attenuation by force measurement setup, and estimation error of the equivalent area from condensers.

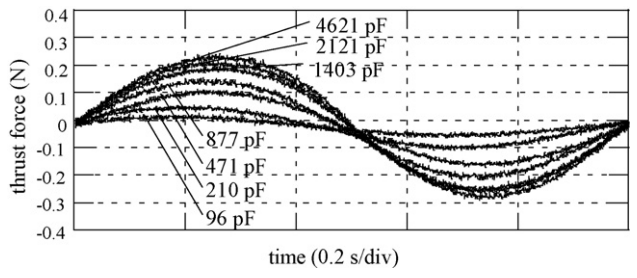


Fig. 20. Thrust force of the motor with capacitance representing induction electrodes.

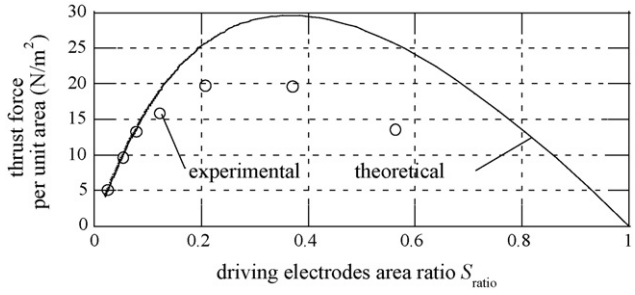


Fig. 21. The variation of thrust force per unit against driving-electrode–area ratio.

7. Experiments of prototype motor

In this section, voltage-induction type film motor is prototyped and its performance is investigated. Thrust force and induced voltage are measured.

7.1. Thrust force of voltage-induction type film motor

We fabricated stator and slider films with induction electrodes as shown in Fig. 22. The stator driving-electrodes and an induction electrode measure 8 mm wide by 30 mm long and 4.5 mm wide by 31.5 mm long, respectively. The slider driving-electrodes and an induction electrode measure 8 mm wide by 15 mm long and 4.5 mm wide by 15 mm long. The driving electrode pitch of the stator is 200 μm and that of the slider is 400 μm.

In the experiments, theoretical values were not calculated by equations in Section 5 but by those in "Appendix A" and following capacitance values, which are based on the model in Fig. 16:

$$C_{ma1} = 1.53\,pF, \quad C_L = 29.5\,pF, \quad C_{SL} = 21.4\,pF, \quad C_i = 17.0\,pF \tag{31}$$

It was because the capacitance between driving electrodes and induction electrodes, which assumed as zero in the Fig. 17, could not be ignored in the prototype motor.

We measured the thrust force with the prototype motor. Although the measurement setup is almost same as Fig. 6, we add two gonio stages to adjust slider tilt because the motor is small and the voltage-induction motor has high sensitivity for tilt. The voltage applied to the driving electrode on the stator was

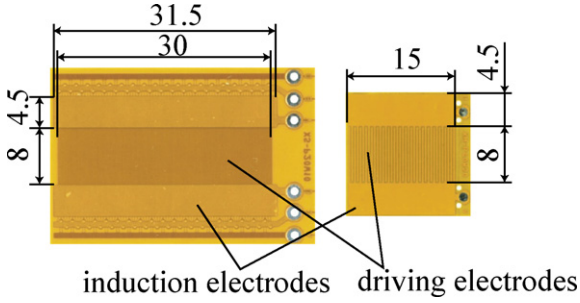


Fig. 22. Voltage-induction type stator (left) and slider (right) films.

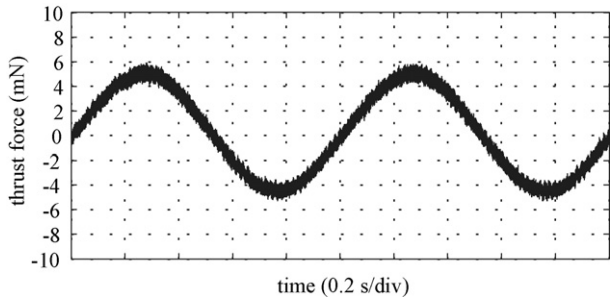


Fig. 23. Thrust force of voltage-induction type electrostatic motor.

$1\,kV_{0-p}$, 1000 Hz. The voltage applied to the induction electrode on the stator was $1\,kV_{0-p}$ and 999 Hz.

Fig. 23 shows the measured thrust force. The main frequency component in the measured thrust force is 1 Hz. That is, the frequency-difference component of the thrust force in (22) appears dominantly. In contrast, the frequency-sum and the frequency-double components are damped and do not appear.

From an FFT analysis, the amplitude of the frequency-difference component was approximately 4.7 mN. The theoretical thrust force value of the component was 7.3 mN.

7.2. Induced voltage of the voltage-induction type film motor

We verified the induction voltage. The voltage applied to the driving electrodes on the stator was $1\,kV_{0-p}$, 1000 Hz and that applied to the induction electrodes on the stator was $1\,kV_{0-p}$, 999 Hz. According to the analysis in (20), the induced voltage can be decomposed to two components, which has angular frequency of ω_t or ω_l. An FFT analysis of the experimental results showed that the amplitude of the induced voltage was 249 V at 999 Hz, and was 51 V at 1000 Hz. The theoretical values for the 999 and 1000 Hz are 297 and 61 V, respectively. The experimental results of the both components becomes 84 percent of the corresponding theoretical values (Fig. 24).

7.3. Motion

Finally, we conducted an experiment to operate the prototype motor. The slider was placed on the stator as in Fig. 25. The voltage to the stator driving electrode and stator induction electrode

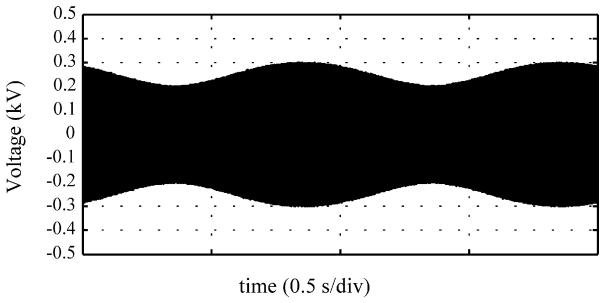


Fig. 24. Induced voltage of voltage-induction type electrostatic motor.

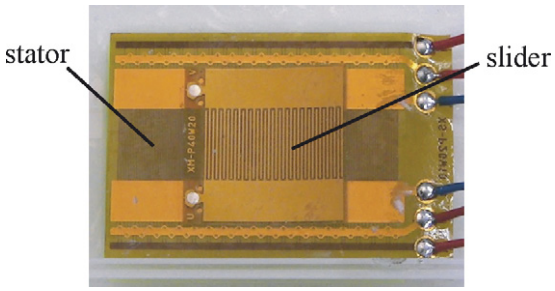


Fig. 25. Voltage-induction type slider is stacked on the stator.

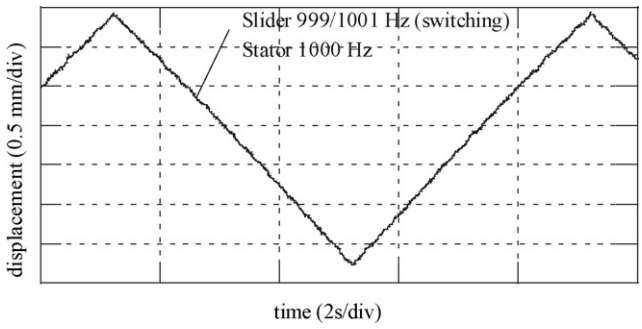


Fig. 26. Voltage-induction type motor motion at the condition of high-frequency voltage.

are $1\,kV_{0-p}$, 1000 Hz, and $1\,kV_{0-p}$, 999/1001 Hz, respectively. The frequency of the two-phase voltage is switched at every two seconds. The motion was measured using the optical displacement sensor (ZIMMER, 200). The result is plotted in Fig. 26. A smooth motion which speed was proportional to the frequency difference was observed. The speed of the slider was 0.8 mm/s, which is same speed as the theoretical analysis predicted. These results verified our analysis and demonstrated the feasibility of the voltage-induction type electrostatic motor.

8. Conclusions

High-power electrostatic motors based on FPC technology are promising for many mechatronic applications because of their high output performance and unique features. Previous high-power motors require electric cabling to their sliders, which may cause troubles in some applications. In this paper, to overcome the problem, we proposed and evaluated a new electrostatic motor using electrostatic induction. The motor has a two- to four-phase driving-electrode configuration, and power feeding to the slider is realized by electrostatic induction, thus removing electric wires to the slider.

The paper first analyzed wired-type of two- to four-phase electrostatic motor ignoring electrostatic induction in order to evaluate the driving-electrode configuration. Based on the capacitance-network model, we revealed that the thrust force generated by this electrode configuration had two frequency components. Then, we proposed suitable operation method of this motor, which uses two high-frequency voltage supplies. The validity of the analysis and operation method was experimentally verified.

Next, the effect of electrostatic induction was analyzed, also based on a capacitance model. We derived the formula of induced voltage, with which thrust force characteristics with induction was calculated. The result of the analysis was verified by experiments with motor model.

Final experiment conducted with a fabricated prototype of the voltage-induction type motor showed that the motor was capable of driving in smooth linear motion with the speed predicted by the analysis.

Although the force of the voltage-induction type motor is lower than that of a wired-type motor, the advantage of no wiring can be useful in many mechatronic applications. In the experimental results provided in this paper, the thrust force is limited to small values since applied voltages were fixed at 1 kV_{0-p} in amplitude. Thrust force can be easily enlarged by increasing the voltage, or by increasing film size, which would be tested in our future work.

Acknowledgments

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Appendix A

In this appendix, we discuss a capacitance-network model that represents a voltage-induction type motor more precisely. In this model, the motor is modeled as an eight-terminal capacitance network as shown in Fig. 16.

A.1. Capacitance-coefficient matrix

$$\mathbf{C}_{\text{all}} = \begin{bmatrix} C_{11} & -C_{12} & -C_{13} & -C_{14} & C_{\text{ma}}(\theta_x) & C_{\text{ma}}(\theta_x-\pi) & -C_{17} & -C_{18} \\ -C_{12} & C_{22} & -C_{23} & -C_{24} & C_{\text{ma}}(\theta_x-\frac{\pi}{2}) & C_{\text{ma}}(\theta_x+\frac{\pi}{2}) & -C_{27} & -C_{28} \\ -C_{13} & -C_{23} & C_{33} & -C_{34} & C_{\text{ma}}(\theta_x-\pi) & C_{\text{ma}}(\theta_x) & -C_{37} & -C_{38} \\ -C_{14} & -C_{24} & -C_{34} & C_{44} & C_{\text{ma}}(\theta_x+\frac{\pi}{2}) & C_{\text{ma}}(\theta_x-\frac{\pi}{2}) & -C_{47} & -C_{48} \\ C_{\text{ma}}(\theta_x) & C_{\text{ma}}(\theta_x-\frac{\pi}{2}) & C_{\text{ma}}(\theta_x-\pi) & C_{\text{ma}}(\theta_x+\frac{\pi}{2}) & C_{\text{SL}}+C_i & -C_L & -C_i & -C_{58} \\ C_{\text{ma}}(\theta_x-\pi) & C_{\text{ma}}(\theta_x+\frac{\pi}{2}) & C_{\text{ma}}(\theta_x) & C_{\text{ma}}(\theta_x-\frac{\pi}{2}) & -C_L & C_{\text{SL}}+C_i & -C_{67} & -C_i \\ -C_{17} & -C_{27} & -C_{37} & -C_{47} & -C_i & -C_{67} & C_{77} & -C_{78} \\ -C_{18} & -C_{28} & -C_{38} & -C_{48} & -C_{58} & -C_i & -C_{78} & C_{88} \end{bmatrix} \tag{32}$$

where $C_{\text{ma}}(\theta_x)$ is represented as follows:

$$C_{\text{ma}}(\theta_x) = -C_{\text{ma0}} - C_{\text{ma1}}\cos(\theta_x) \tag{33}$$

A.2. Induced voltages

Using (32), (17), and (19), we can represent the relation of the electrodes as follows:

$$\mathbf{q}_I = \mathbf{C}_{\text{all}}\mathbf{V}_I \tag{34}$$

Then, following formulas of induced voltages are derived:

$$V_5 = \frac{1}{C_i + C_L + C_{\text{SL}}}\left\{\left(C_i - C_{58} + \frac{-C_{58}C_L + C_{67}C_L}{C_i - C_L + C_{\text{SL}}}\right) \times V_l \sin\omega_l t + 2C_{\text{ma1}}V_t \sin(\omega_t t - \theta_x)\right\}$$
$$V_6 = -V_5 \tag{35}$$

Comparing with (20), we can find that this formula also has two frequency components.

A.3. Thrust force

As same way in (21), we can derive the thrust force formula as follows:

$$f_{\text{IND}} = \frac{\pi C_{\text{ma1}}V_t}{2p(C_i + C_L + C_{\text{SL}})}\{4C_{\text{ma1}}V_t \sin(2\omega_t t - 2\theta_x) + (2C_i - C_{58} - C_{67})V_l \sin(\omega_l t + \omega_t t - \theta_x) + (2C_i - C_{58} - C_{67})V_l \sin(\omega_l t - \omega_t t + \theta_x)\} \tag{36}$$

This equation also has three frequency components as same as (22), that is, the frequencies are the double frequency of stator driving voltage, the sum of frequencies, and the difference of frequencies.

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